VISVESVARAYA TECHNOLOGICAL UNIVERSITY

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LAB REPORT

On

ANALYSIS AND DESIGN OF ALGORITHMS (23CS4PCADA)

Submitted by

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in partial fulfillment for the award of the degree of
BACHELOR OF ENGINEERING
in
COMPUTER SCIENCE AND ENGINEERING

B.M.S. COLLEGE OF ENGINEERING
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CERTIFICATE

B. M. S. College of Engineering,
Bull Temple Road, Bangalore 560019
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Department of Computer Science and Engineering

This is to certify that the Lab work entitled **"ANALYSIS AND DESIGN OF ALGORITHMS"** carried out by AYUSH S SHETTY**(1BM24CS064)**, who is bonafide student of **B. M. S. College of Engineering**. It is in partial fulfillment for the award of **Bachelor of Engineering in Computer Science and Engineering** of the Visvesvaraya Technological University, Belgaum during the year 2025-26. The Lab report has been approved as it satisfies the academic requirements in respect of Analysis and Design of Algorithms Lab - **(23CS4PCADA)** work prescribed for the said degree.

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Index Sheet

Course outcomes:

CO1	Analyze time complexity of algorithms using asymptotic notations.
CO2	Apply various algorithm design techniques for the given problem.
CO3	Evaluate the appropriate algorithm/design technique for solving a given problem.
CO4	Conduct practical experiments by applying appropriate algorithm design technique to solve problems.

Lab program 1:

Write a program to obtain the Topological ordering of vertices in a given digraph.

C Code:

```
#include <stdio.h>
#define MAX 10
int main() {
   int n, i, j, graph[MAX][MAX], indegree[MAX] = {0};
   int queue[MAX], front = 0, rear = -1, topo[MAX], k = 0;

   printf("Enter number of vertices: ");
   scanf("%d", &n);
   printf("Enter adjacency matrix:\n");
   for (i = 0; i < n; i++) {
      for (j = 0; j < n; j++) {
         scanf("%d", &graph[i][j]);
         if (graph[i][j] == 1) indegree[j]++;
      }
   }
   for (i = 0; i < n; i++) if (indegree[i] == 0) queue[++rear] = i;
   while (front <= rear) {
      int v = queue[front++]; topo[k++] = v;
      for (j = 0; j < n; j++) {
         if (graph[v][j] == 1) {
            indegree[j]--;
            if (indegree[j] == 0) queue[++rear] = j;
         }
      }
   }
   printf("Topological Order: ");
   for (i = 0; i < k; i++) printf("%d ", topo[i]);
   return 0;
}
```

OUTPUT:

```
Enter number of vertices: 3
Enter adjacency matrix:
0 1 0
0 0 1
0 0 0
Topological Order: 0 1 2
Process returned 0 (0x0)   execution time : 28.434 s
Press any key to continue.
```

Lab program 2:

Implement Johnson Trotter algorithm to generate permutations.

C Code:

```
#include <stdio.h>

int getMobile(int a[], int dir[], int n)
{
   int mobile = 0, mobile_pos = -1;

   for(int i = 0; i < n; i++)
   {
      if(dir[i] == -1 && i != 0 && a[i] > a[i-1])   // Left
      {
         if(a[i] > mobile)
         {
            mobile = a[i];
            mobile_pos = i;
         }
      }

      if(dir[i] == 1 && i != n-1 && a[i] > a[i+1])  // Right
      {
         if(a[i] > mobile)
         {
            mobile = a[i];
            mobile_pos = i;
         }
      }
   }

   return mobile_pos;
}

void printPermutation(int a[], int n)
{
   for(int i = 0; i < n; i++)
      printf("%d ", a[i]);
   printf("\n");
}

int main()
{
   int n;

   printf("Enter n: ");
   scanf("%d", &n);
```

```
int a[n], dir[n];

// Initial permutation
for(int i = 0; i < n; i++)
{
   a[i] = i + 1;
   dir[i] = -1;   // -1 = Left, 1 = Right
}

printf("Permutations:\n");
printPermutation(a, n);

while(1)
{
   int pos = getMobile(a, dir, n);

   if(pos == -1)
      break;

   int mobile = a[pos];

   // Swap mobile element
   if(dir[pos] == -1)
   {
      int temp = a[pos];
      a[pos] = a[pos-1];
      a[pos-1] = temp;

      temp = dir[pos];
      dir[pos] = dir[pos-1];
      dir[pos-1] = temp;

      pos--;
   }
   else
   {
      int temp = a[pos];
      a[pos] = a[pos+1];
      a[pos+1] = temp;

      temp = dir[pos];
      dir[pos] = dir[pos+1];
      dir[pos+1] = temp;

      pos++;
   }

   // Reverse directions of elements larger than mobile
   for(int i = 0; i < n; i++)
```

```
        {
            if(a[i] > mobile)
                dir[i] = -dir[i];
        }

        printPermutation(a, n);
    }

    return 0;
}
```

OUTPUT:

```
Enter the number of elements (n): 3

Starting Permutation:
1<- 2<- 3<-
1<- 3<- 2<-
3<- 1<- 2<-
3-> 2<- 1<-
2<- 3-> 1<-
2<- 1<- 3->

Process returned 0 (0x0)   execution time : 48.162 s
Press any key to continue.
```

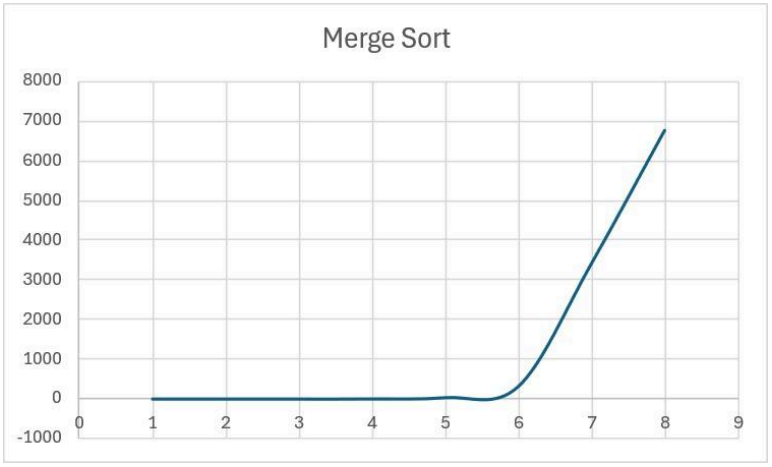
Lab Program 3:

Sort a given set of N integer elements using Merge Sort technique and compute its time taken. Run the program for different values of N and record the time taken to sort.

C Code:

```
#include <stdio.h>

void merge(int arr[], int l, int m, int r)
{
   int i, j, k;
   int n1 = m - l + 1;
   int n2 = r - m;

   int L[n1], R[n2];

   for (i = 0; i < n1; i++)
      L[i] = arr[l + i];
   for (j = 0; j < n2; j++)
      R[j] = arr[m + 1 + j];
   i = 0;
   j = 0;
   k = l;


   while (i < n1 && j < n2)
   {
      if (L[i] <= R[j])
      {
         arr[k] = L[i];
         i++;
      }
      else
      {
         arr[k] = R[j];
         j++;
      }
      k++;
   }
   while (i < n1)
   {
      arr[k] = L[i];
      i++;
      k++;
   }
   while (j < n2)
   {
      arr[k] = R[j];
```

```
        j++;
        k++;
    }
}

void mergesort(int arr[], int l, int r)
{
    if (l < r)
    {
        int m = l + (r - l) / 2;
        mergesort(arr, l, m);
        mergesort(arr, m + 1, r);
        merge(arr, l, m, r);
    }
}

int main()
{
    int n, i;
    printf("Enter the number of elements: ");
    scanf("%d", &n);
    int a[n];
    printf("Enter the elements: ");
    for (i = 0; i < n; i++)
    {
        scanf("%d", &a[i]);
    }
    mergesort(a, 0, n - 1);
    printf("Sorted array: ");
    for (i = 0; i < n; i++)
    {
        printf("%d ", a[i]);
    }
    printf("\n");
    return 0;
}
```



OUTPUT:

```
Enter the number of elements: 5
Enter the elements: 32 34 54 21 32
Sorted array: 21 32 32 34 54

Process returned 0 (0x0)   execution time : 14.935 s
Press any key to continue.
```

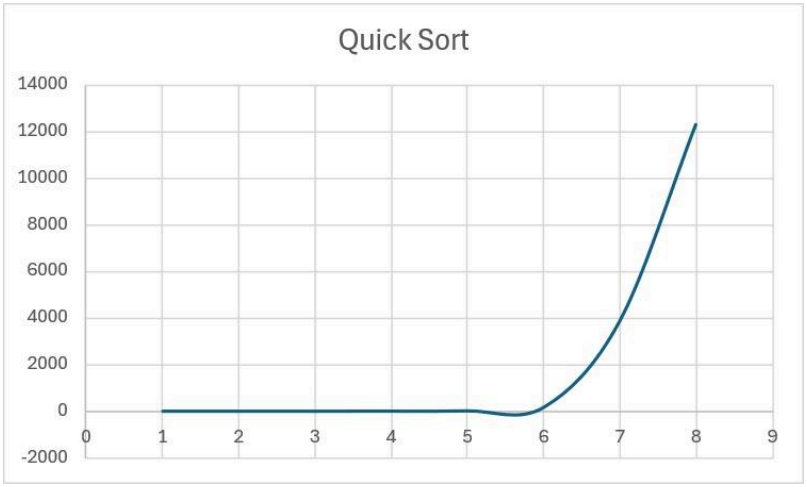
Lab Program 4:

Sort a given set of N integer elements using Quick Sort technique and compute its time taken.

C Code:

```
#include <stdio.h>

int partition(int a[], int low, int high) {
   int key = a[low];
   int i = low + 1;
   int j = high;
   int temp;

   while (i <= j) {
      while (i <= high && a[i] <= key) {
         i++;
      }
      while (a[j] > key) {
         j--;
      }
      if (i < j) {
         temp = a[i];
         a[i] = a[j];
         a[j] = temp;
      }
   }

   temp = a[low];
   a[low] = a[j];
   a[j] = temp;
   return j;
}
```

```
void quickSort(int a[], int low, int high) {
    if (low < high) {
        int pi = partition(a, low, high);
        quickSort(a, low, pi - 1);
        quickSort(a, pi + 1, high);
    }
}

int main() {
    int n, i;
    printf("Enter the number of elements: ");
    scanf("%d", &n);
    int a[n];
    printf("Enter the elements: ");
    for (i = 0; i < n; i++) {
        scanf("%d", &a[i]);
    }

    quickSort(a, 0, n - 1);

    printf("Sorted array: ");
    for (i = 0; i < n; i++) {
        printf("%d ", a[i]);
    }
    printf("\n");

    return 0;
}
```



Output:

```
Enter the number of elements: 7
Enter the elements: 11 23 43 56 76 54 67 87 98
Sorted array: 11 23 43 54 56 67 76

Process returned 0 (0x0)   execution time : 20.961 s
Press any key to continue.
```

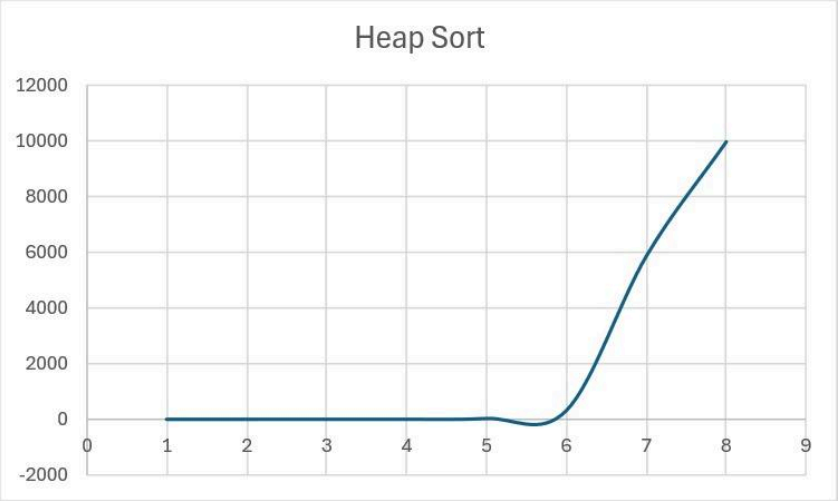
Lab Program 5:

Sort a given set of N integer elements using Heap Sort technique and compute its time taken.

C Code:

```
#include <stdio.h>

void heapify(int arr[], int n, int i) {
    int largest = i;
    int l = 2 * i + 1;
    int r = 2 * i + 2;

    if (l < n && arr[l] > arr[largest])
        largest = l;

    if (r < n && arr[r] > arr[largest])
        largest = r;

    if (largest != i) {
        int temp = arr[i];
        arr[i] = arr[largest];
        arr[largest] = temp;
        heapify(arr, n, largest);
    }
}

void heapSort(int arr[], int n) {
    for (int i = n / 2 - 1; i >= 0; i--)
        heapify(arr, n, i);

    for (int i = n - 1; i > 0; i--) {
        int temp = arr[0];
        arr[0] = arr[i];
        arr[i] = temp;
        heapify(arr, i, 0);
    }
}
```

```
int main() {
    int n, i;
    printf("Enter the number of elements: ");
    scanf("%d", &n);
    int a[n];
    printf("Enter the elements: ");
    for (i = 0; i < n; i++) {
        scanf("%d", &a[i]);
    }

    heapSort(a, n);

    printf("Sorted array: ");
    for (i = 0; i < n; i++) {
        printf("%d ", a[i]);
    }
    printf("\n");

    return 0;
}
```



Output:

```
Enter the number of elements: 8
Enter the elements: 9 4 3 8 10 2 5 7
Sorted array: 2 3 4 5 7 8 9 10

Process returned 0 (0x0)   execution time : 35.063 s
Press any key to continue.
```

Lab Program 6:

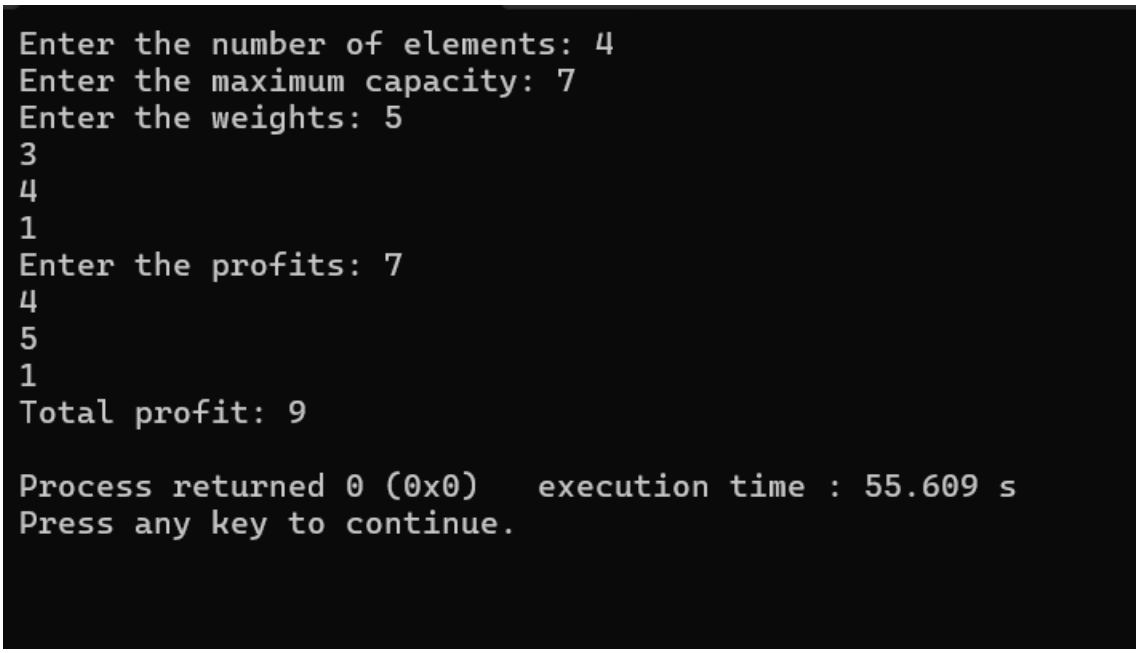
Implement 0/1 Knapsack problem using dynamic programming.

C Code:

```
#include <stdio.h>

int main()
{
   int n, w[50], p[50], m, a[50][50];
   int i, j;
   printf("Enter the number of elements: ");
   scanf("%d", &n);
   printf("Enter the maximum capacity: ");
   scanf("%d", &m);
   printf("Enter the weights: ");
   for (i = 1; i <= n; i++)
   {
      scanf("%d", &w[i]);
   }
   printf("Enter the profits: ");
   for (i = 1; i <= n; i++)
   {
      scanf("%d", &p[i]);
   }
   for (i = 0; i <= n; i++)
      a[i][0] = 0;
   for (j = 0; j <= m; j++)
      a[0][j] = 0;
   for (i = 1; i <= n; i++)
   {
      for (j = 1; j <= m; j++)
      {
         if (j < w[i])
         {
            a[i][j] = a[i - 1][j];
         }
         else
         {
            if (a[i - 1][j] > (p[i] + a[i - 1][j - w[i]]))
            {
               a[i][j] = a[i - 1][j];
            }
            else
            {
               a[i][j] = p[i] + a[i - 1][j - w[i]];
            }
         }
```

```
        }
    }
    printf("Total profit: %d\n", a[n][m]);
    return 0;
}
```

Output:



Lab Program 7:

Implement All Pair Shortest paths problem using Floyd's algorithm.

C Code:

```
#include <stdio.h>

void Floyds(int n, int cost[50][50], int A[50][50])
{
   int i, j, k;

   for(i = 1; i <= n; i++)
      for(j = 1; j <= n; j++)
         A[i][j] = cost[i][j];

   for(k = 1; k <= n; k++)
      for(i = 1; i <= n; i++)
         for(j = 1; j <= n; j++)
            if(A[i][k] + A[k][j] < A[i][j])
               A[i][j] = A[i][k] + A[k][j];
}

int main() {
   int n, i, j;
   int cost[50][50], A[50][50];
   printf("Enter number of vertices: ");
   scanf("%d", &n);
   printf("Enter the cost matrix (use 999 for infinity):\n");
   for(i = 1; i <= n; i++) {
      for(j = 1; j <= n; j++) {
         scanf("%d", &cost[i][j]);
         if (cost[i][j] == 0 && i != j) {
            cost[i][j] = 999;
         }
      }
   }
   Floyds(n, cost, A);
   printf("\nThe all-pairs shortest path matrix is:\n");
   for(i = 1; i <= n; i++) {
      for(j = 1; j <= n; j++) {
         if (A[i][j] == 999) printf("INF ");
         else printf("%d  ", A[i][j]);
      }
      printf("\n");
   }

   return 0;
}
```

Output:

```
Enter number of vertices: 4
Enter the cost matrix (use 999 for infinity):
0 999 6 1
4 0 20 10
999 3 0 12
6 999 999 0

The all-pairs shortest path matrix is:
0  9  6  1
4  0  10  5
7  3  0  8
6  15  12  0

Process returned 0 (0x0)   execution time : 54.621 s
Press any key to continue.
```

Lab Program 8a:

Find Minimum Cost Spanning Tree of a given undirected graph using Prim's algorithm.

C Code:

```
#include <stdio.h>


int main()
{
   int n, cost[10][10];
   int visited[10] = {0};
   int min, u, v, edges = 0;
   int mincost = 0;

   printf("Enter number of vertices: ");
   scanf("%d", &n);

   printf("Enter cost matrix:\n");
   for(int i = 0; i < n; i++)
   {
      for(int j = 0; j < n; j++)
      {
         scanf("%d", &cost[i][j]);

         if(cost[i][j] == 0)
            cost[i][j] = 999;   // No edge
      }
   }
```

```
visited[0] = 1;  // Start from vertex 0

printf("\nEdges in MST:\n");

while(edges < n - 1)
{
   min = 999;

   for(int i = 0; i < n; i++)
   {
      if(visited[i])
      {
         for(int j = 0; j < n; j++)
         {
            if(!visited[j] && cost[i][j] < min)
            {
               min = cost[i][j];
               u = i;
               v = j;
            }
         }
      }
   }

   printf("%d -> %d : %d\n", u, v, min);
```

```
        mincost += min;
        visited[v] = 1;
        edges++;
    }

    printf("\nTotal minimum Cost : %d\n", mincost);

    return 0;
}
```

Output:

```
Enter no of vertices: 4
Enter Adjacency matrix (use 999 for no edge):
0 2 999 6
2 0 3 8
999 3 0 0
6 8 0 0

Edge    Weight
0 - 1    2
1 - 2    3
0 - 3    6

Total minimum Cost: 11
```

Lab Program 8 b:

Find Minimum Cost Spanning Tree of a given undirected graph using Kruskal's algorithm.

C Code:

```
#include <stdio.h>

int parent[10];

int find(int v)
{
   while(parent[v] != 0)
      v = parent[v];
   return v;
}

void unionSet(int u, int v)
{
   parent[v] = u;
}

int main()
{
   int n, cost[10][10];
   int mincost = 0;
   int edges = 0;

   printf("Enter number of vertices: ");
   scanf("%d", &n);

   printf("Enter cost matrix:\n");
   for(int i = 0; i < n; i++)
   {
      for(int j = 0; j < n; j++)
      {
         scanf("%d", &cost[i][j]);

         if(cost[i][j] == 0)
            cost[i][j] = 999;
      }
   }

   printf("\nEdges in MST:\n");

   while(edges < n - 1)
   {
      int min = 999;
      int u = -1, v = -1;
```

```
        // Find minimum edge
        for(int i = 0; i < n; i++)
        {
            for(int j = 0; j < n; j++)
            {
                if(cost[i][j] < min)
                {
                    min = cost[i][j];
                    u = i;
                    v = j;
                }
            }
        }

        int p = find(u);
        int q = find(v);

        // If no cycle
        if(p != q)
        {
            printf("%d -> %d : %d\n", u, v, min);
            mincost += min;
            edges++;
            unionSet(p, q);
        }

        cost[u][v] = cost[v][u] = 999;
    }

    printf("\nMinimum Cost = %d\n", mincost);

    return 0;
}
```

Output:

```
Enter the number of edges: 5
Enter src, dest and cost for following
0 1 10
0 2 6
0 3 5
1 3 15
2 3 4

For the total cost is: 19

Process returned 0 (0x0)   execution time : 39.230 s
Press any key to continue.
```

Lab Program 9:

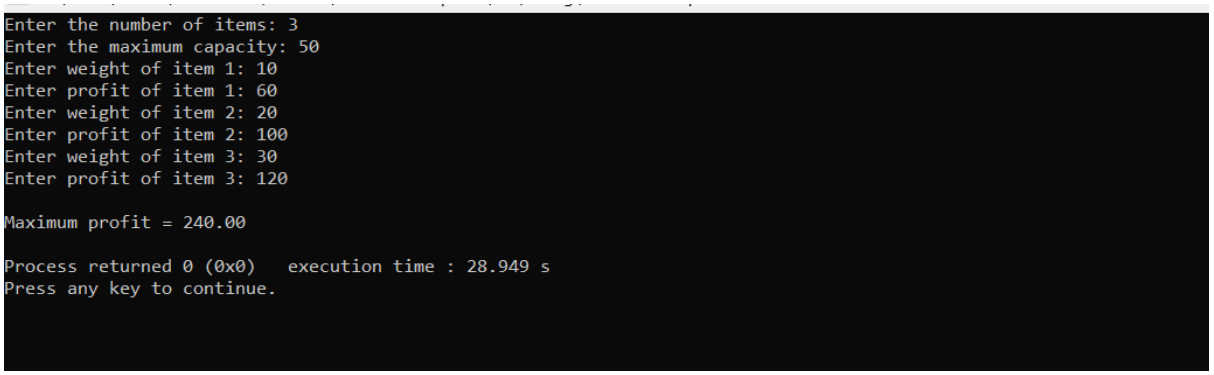
Implement fractional knapsack problem using Greedy technique.

C Code:

```
#include<stdio.h>
struct item{
   int p;
   int w;
   float r;
   float x;
};
int main(){
   int n, m;
   printf("Enter the number of items: ");
   scanf("%d", &n);
   printf("Enter the maximum capacity: ");
   scanf("%d", &m);
   struct item items[20], temp;
   for(int i=0; i<n; i++){
      printf("Enter weight of item %d: ", i+1);
      scanf("%d", &items[i].w);
      printf("Enter profit of item %d: ", i+1);
      scanf("%d", &items[i].p);
      items[i].x = 0;
      items[i].r = (float)items[i].p / items[i].w;
   }
   for(int i=0; i<n; i++){
      for(int j=i+1; j<n; j++){
         if(items[j].r > items[i].r){
            temp = items[i];
            items[i] = items[j];
            items[j] = temp;
         }
      }
   }
   float remain = m, maxprofit = 0;
   for(int i=0; i<n; i++){
      if(items[i].w <= remain){
         items[i].x = 1;
         remain -= items[i].w;
         maxprofit += items[i].p;
      } else {
         items[i].x = remain / items[i].w;
         maxprofit += items[i].p * items[i].x;
         break;
         }
   }
```

```
    printf("\nMaximum profit = %.2f\n", maxprofit);
}
```

Output:



Lab Program 10:

From a given vertex in a weighted connected graph, find shortest paths to other vertices using Dijkstra's algorithm.

C Code

```
#include <stdio.h>

int main()
{
   int n, cost[10][10];
   int dist[10], visited[10];
   int source;

   printf("Enter number of vertices: ");
   scanf("%d", &n);

   printf("Enter cost matrix:\n");
   for(int i = 0; i < n; i++)
   {
      for(int j = 0; j < n; j++)
      {
         scanf("%d", &cost[i][j]);

         if(i != j && cost[i][j] == 0)
            cost[i][j] = 999;
      }
   }

   printf("Enter source vertex: ");
   scanf("%d", &source);

   for(int i = 0; i < n; i++)
   {
      dist[i] = cost[source][i];
      visited[i] = 0;
   }

   dist[source] = 0;
   visited[source] = 1;

   for(int count = 1; count < n; count++)
   {
      int min = 999, u = -1;

      // Find nearest unvisited vertex
      for(int i = 0; i < n; i++)
      {
         if(!visited[i] && dist[i] < min)
```

```
            {
                min = dist[i];
                u = i;
            }
        }

        visited[u] = 1;

        // Relax adjacent vertices
        for(int v = 0; v < n; v++)
        {
            if(!visited[v] &&
               dist[u] + cost[u][v] < dist[v])
            {
                dist[v] = dist[u] + cost[u][v];
            }
        }
    }

    printf("\nShortest distances from vertex %d:\n", source);

    for(int i = 0; i < n; i++)
    {
        printf("%d -> %d = %d\n", source, i, dist[i]);
    }

    return 0;
}
```

Output:

```
Enter number of vertices
4
Enter the cost matrix
0 2 0 6
2 0 3 8
0 3 0 0
6 8 0 0
Enter source vertex
0
Shortest distances from vertex 0:
to vertex 1 = 5
to vertex 2 = 3
to vertex 3 = 0
to vertex 4 = 11
```

Lab Program 11:

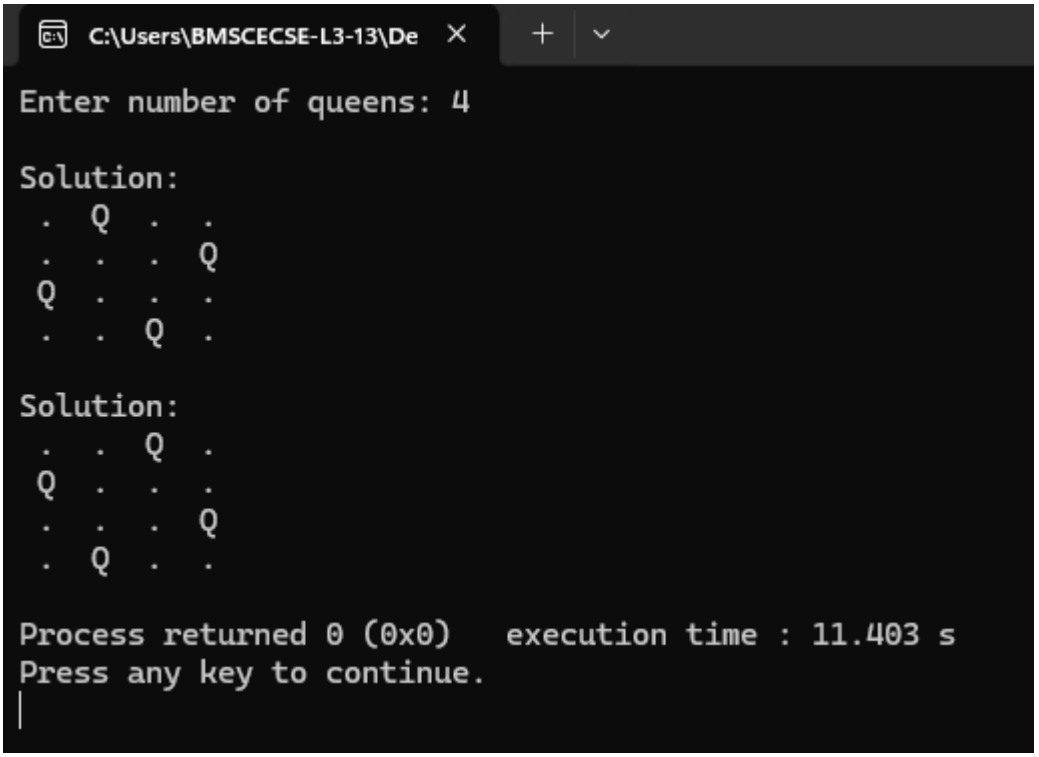
Implement “N-Queens Problem” using Backtracking.

C Code:

```
#include <stdio.h>
#include <stdlib.h>
#define MAX 20
int x[MAX];
int PLACE(int k, int i)
{
   int j;
   for (j = 1; j <= k - 1; j++)
   {
      if ((x[j] == i) || (abs(x[j] - i) == abs(j - k)))
         return 0;
   }
   return 1;
}

void printSolution(int n)
{
   int i, j;
   printf("\nSolution:\n");
   for (i = 1; i <= n; i++)
   {
      for (j = 1; j <= n; j++)
      {
         if (x[i] == j)
            printf(" Q ");
         else
            printf(" . ");
      }
      printf("\n");
   }
}

void NQueens(int n)
{
   int k = 1;
   x[k] = 0;
   while (k != 0)
   {
      x[k] = x[k] + 1;
      while ((x[k] <= n) && !PLACE(k, x[k]))
      {
         x[k] = x[k] + 1;
      }
      if (x[k] <= n)
```

```
        {
            if (k == n)
            {
                printSolution(n);
            }
            else
            {
                k++;
                x[k] = 0;
            }
        }
        else
        {
            k--;
        }
    }
}

int main()
{
    int n;
    printf("Enter number of queens: ");
    scanf("%d", &n);
    NQueens(n);
}
```

Output:

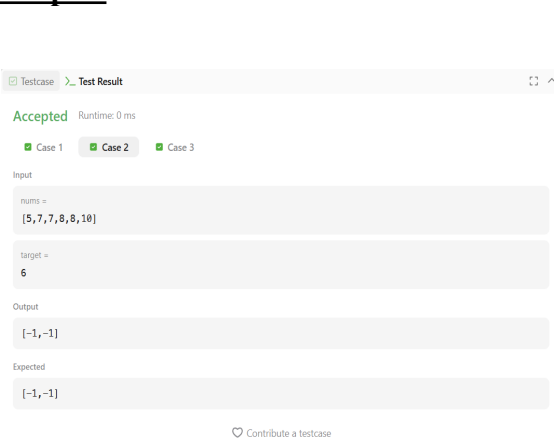


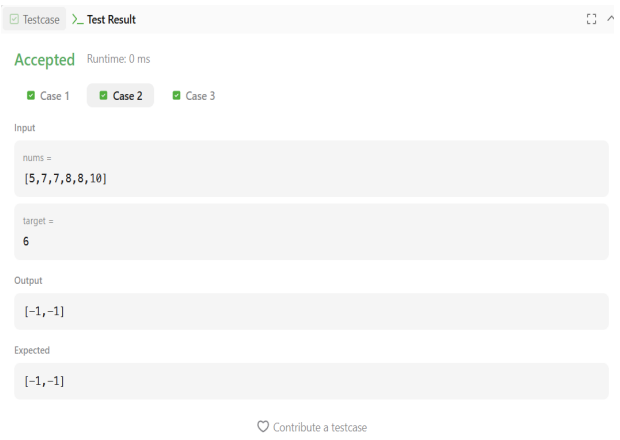
LEETCODE 34. Find First and Last Position of Element in Sorted Array

Program:

```
#include <stdlib.h>

int* searchRange(int* nums, int numsSize, int target, int* returnSize) {
    *returnSize = 2;

    int* result = (int*)malloc(2 * sizeof(int));
    result[0] = -1;
    result[1] = -1;

    for (int i = 0; i < numsSize; i++) {
        if (nums[i] == target) {
            result[0] = i;
            break;
        }
    }
    for (int i = numsSize - 1; i >= 0; i--) {
        if (nums[i] == target) {
            result[1] = i;
            break;
        }
    }
    return result;
}
```

Output:



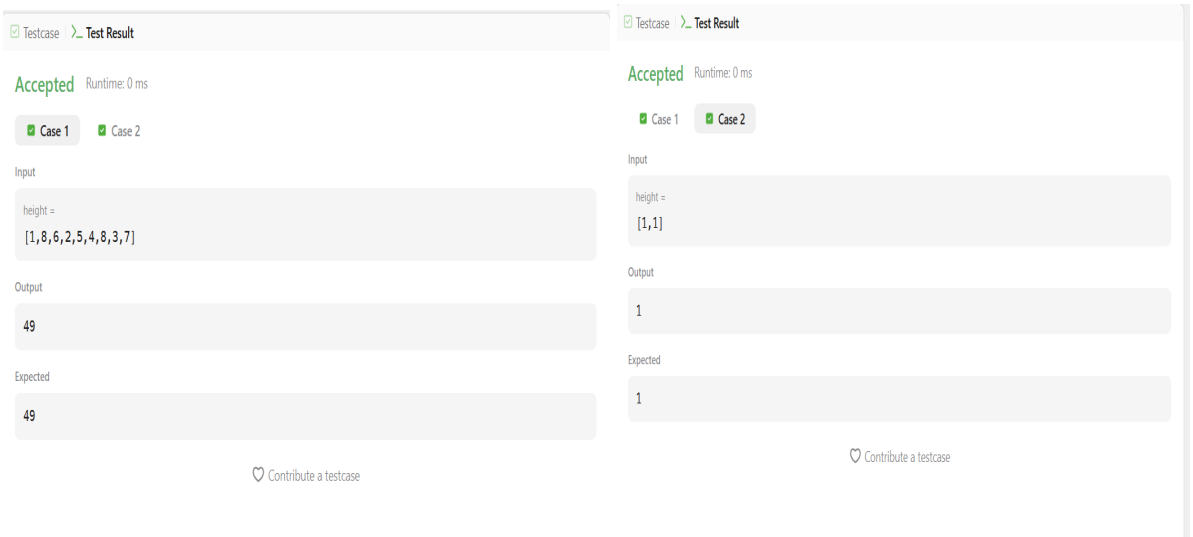


LEETCODE 11. Container With Most Water

Program:

```
int maxArea(int* height, int heightSize) {
    int i=0;
    int j=heightSize-1;
    int res=0;
    while(i<j){
        int minH=height[i]<height[j]?height[i]:height[j];
        int area=(j-i)*minH;
        if(area>res){
            res=area;
        }
        if(height[i]<height[j]){
            i++;
        }
        else{
            j--;
        }
    }return res;
}
```

Output:

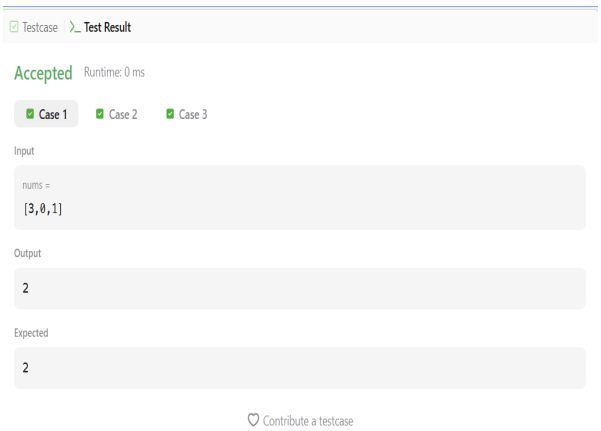


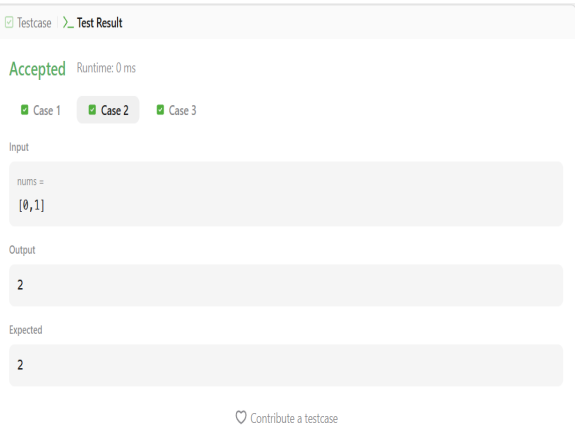
LEETCODE 268. Missing Number

Program:

```
int missingNumber(int* nums, int numsSize) {
    int i, j;

    for(i = 0; i < numsSize - 1; i++) {
        for(j = 0; j < numsSize - i - 1; j++) {
            if(nums[j] > nums[j + 1]) {
                int temp = nums[j];
                nums[j] = nums[j + 1];
                nums[j + 1] = temp;
            }
        }
    }

    for(i = 0; i < numsSize; i++) {
        if(nums[i] != i) {
            return i;
    }

    return numsSize;
}
```

Output:



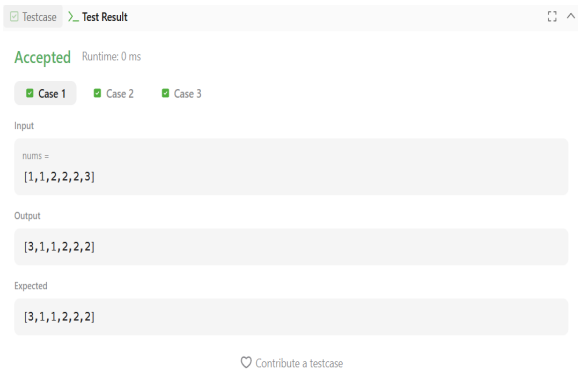


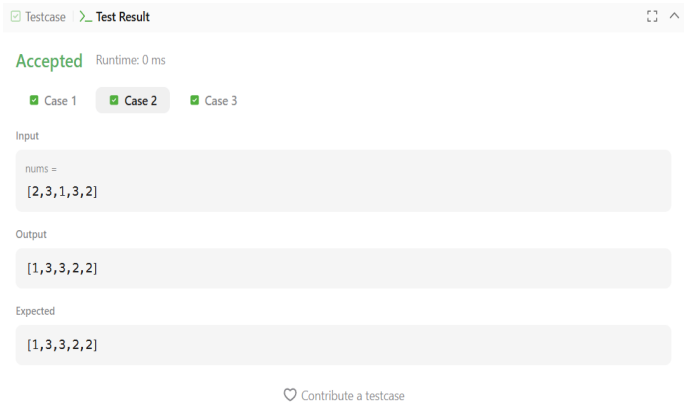
LEETCODE 1636. Sort Array by Increasing Frequency

Program:

```
int* frequencySort(int* nums, int numsSize, int* returnSize) {
    *returnSize = numsSize;
    int i, j, k;
    for (i = 0; i < numsSize - 1; i++) {
        for (j = 0; j < numsSize - i - 1; j++) {
            int f1 = 0, f2 = 0;
            for (k = 0; k < numsSize; k++) {
                if (nums[k] == nums[j])   f1++;
                if (nums[k] == nums[j+1]) f2++;
            }
            if (f1 > f2 || (f1 == f2 && nums[j] < nums[j+1])) {
                int temp = nums[j];
                nums[j] = nums[j+1];
                nums[j+1] = temp;
            }
        }
    }
    return nums;
}
```

Output:



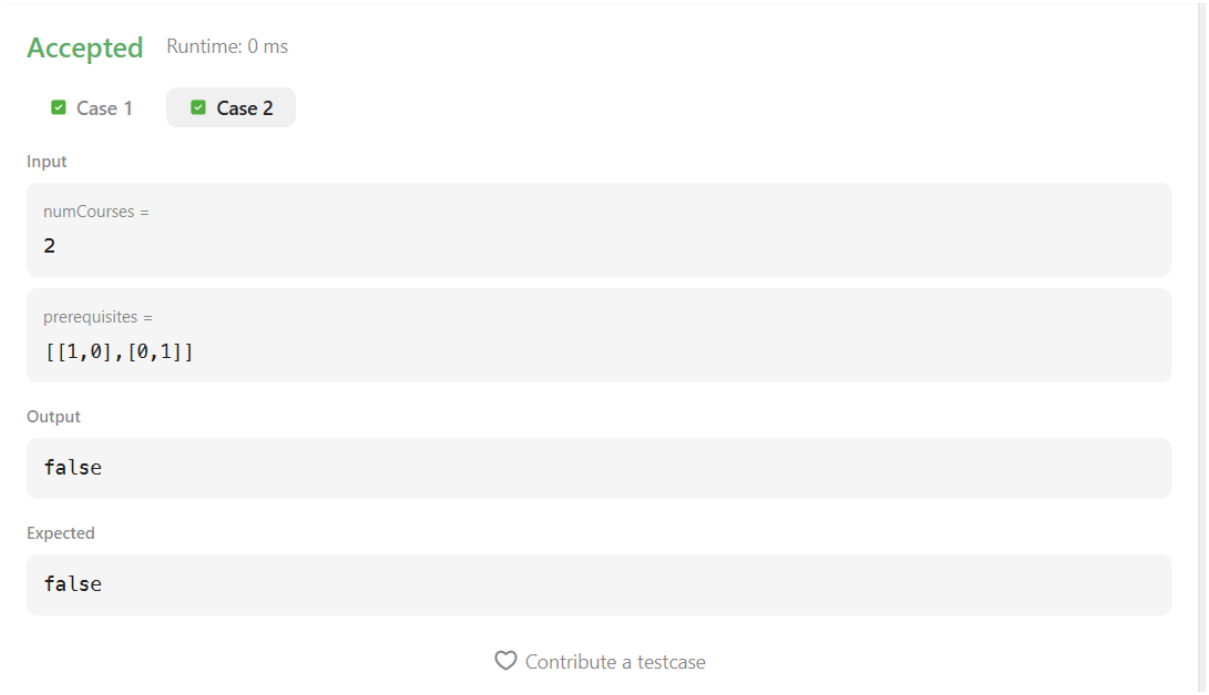


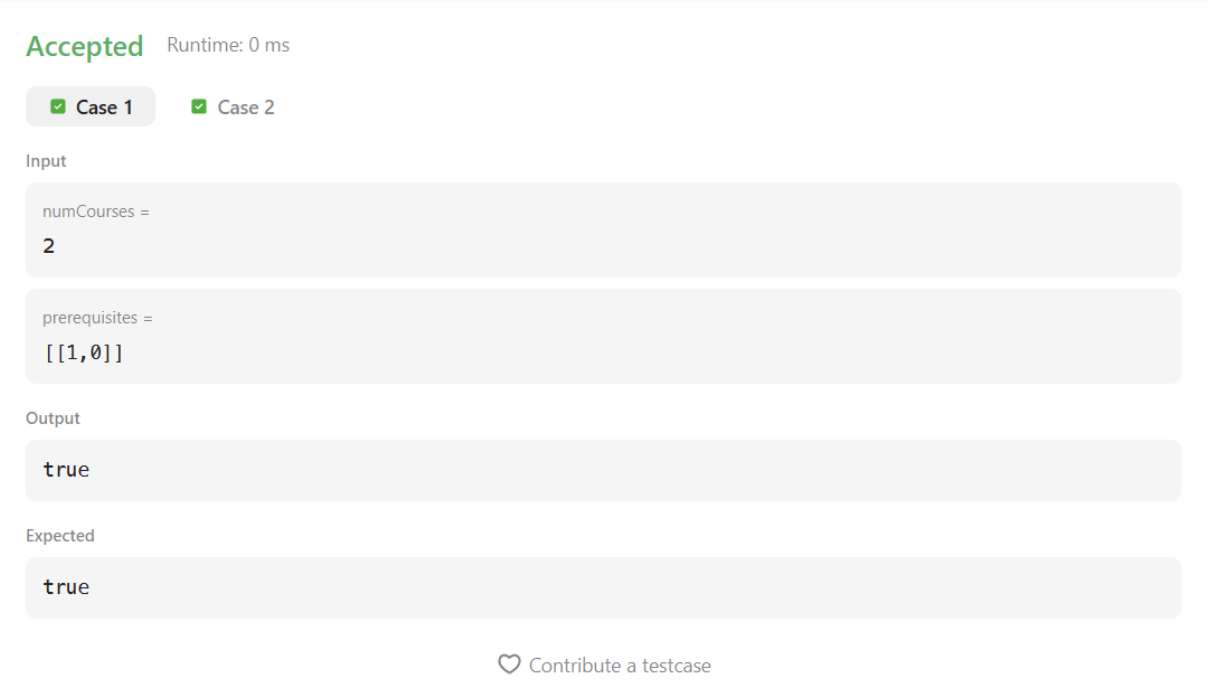
LEETCODE 207. Course Schedule

Program:

```
bool canFinish(int numCourses, int** prerequisites, int prerequisitesSize, int*
prerequisitesColSize) {
    int *indegree = calloc(numCourses, sizeof(int));
    int *outDegree = calloc(numCourses, sizeof(int));
    for (int i = 0; i < prerequisitesSize; i++) {
        outDegree[prerequisites[i][1]]++;
        indegree[prerequisites[i][0]]++;
    }
    int **graph = malloc(numCourses * sizeof(int *));
    for (int i = 0; i < numCourses; i++) {
        graph[i] = malloc(outDegree[i] * sizeof(int));
        outDegree[i] = 0;
    }
    for (int i = 0; i < prerequisitesSize; i++) {
        int u = prerequisites[i][1];
        int v = prerequisites[i][0];
        graph[u][outDegree[u]++] = v;
    }
    int *queue = malloc(numCourses * sizeof(int));
    int front = 0, rear = 0;
    for (int i = 0; i < numCourses; i++) {
        if (indegree[i] == 0) {
            queue[rear++] = i;
        }
    }
    int count = 0;
    while (front < rear) {
        int u = queue[front++];
        count++;
        for (int i = 0; i < outDegree[u]; i++) {
            int v = graph[u][i];
            if (--indegree[v] == 0) {
                queue[rear++] = v;
            }
        }
    }
    for (int i = 0; i < numCourses; i++) {
        free(graph[i]);
    }
    free(graph);
    free(indegree);
    free(outDegree);
    free(queue);
    return count == numCourses;
}
```

Output:


Accepted
Runtime: 0 ms
Case 1
Case 2
Input
numCourses =
2
prerequisites =
[[1,0],[0,1]]
Output
false
Expected
false
Contribute a testcase

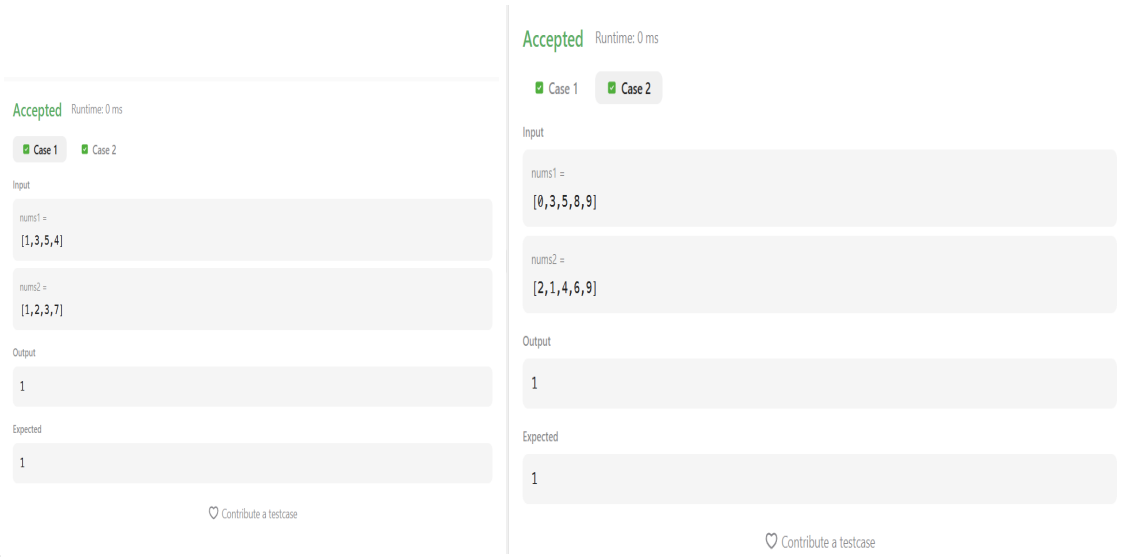

Accepted
Runtime: 0 ms
Case 1
Case 2
Input
numCourses =
2
prerequisites =
[[1,0]]
Output
true
Expected
true
Contribute a testcase

LEETCODE 801. Minimum Swaps To Make Sequences Increasing

Program:

```
int minSwap(int* nums1, int nums1Size, int* nums2, int nums2Size) {
    int keep = 0;
    int swap = 1;
    for (int i = 1; i < nums1Size; i++) {
        int nextKeep = nums1Size + 1;
        int nextSwap = nums1Size + 1;
        if (nums1[i] > nums1[i - 1] && nums2[i] > nums2[i - 1]) {
            nextKeep = keep;
            nextSwap = swap + 1;
        }
        if (nums1[i] > nums2[i - 1] && nums2[i] > nums1[i - 1]) {
            if (swap < nextKeep)
                nextKeep = swap;
            if (keep + 1 < nextSwap)
                nextSwap = keep + 1;
        }
        keep = nextKeep;
        swap = nextSwap;
    }
    return keep < swap ? keep : swap;
}
```

Output:



-

LEETCODE 287. Find the Duplicate Number

Program:

```
int findDuplicate(int* nums, int numsSize) {

    int slow = nums[0];
    int fast = nums[0];


    do{
        slow = nums[slow];
        fast = nums[nums[fast]];
    }while (slow !=fast);

    slow = nums[0];
    while (slow !=fast){
        slow = nums[slow];
        fast = nums[fast];
    }

     return slow;
}
```

Output: